Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/13/2023
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Alteration of Bacillus subtilis DNA Architecture in Space: Global Effects on DNA Supercoiling, Methylation, and the Transcriptome		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	12/03/2020	**End Date:**	12/02/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: This project has been transitioned to Dr. Kelly Rice, due to the death of Dr. Nicholson. Dr. Rice became Principal Investigator for this project August 21, 2023. Please see the project under Dr. Rice for continued progress (Ed., 11/20/23).		
COI Name (Institution):	Fajardo-Cavazos, Patricia Ph.D. (University of Florida, Gainesville) Wang, Xindan Ph.D. (Indiana University)		
Grant/Contract No.:	80NSSC21K0319		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	1) Description of the research product(s): Studying the response of microbial cells to the spaceflight environment is a high-priority issue in Space Biology, due to the importance of microbes in maintaining the health of astronauts and their habitats. Previous International Space Station (ISS) flight experiments (missions Biological Research in Canisters (BRIC)-18, -21, and -23)) using the Gram-positive model bacterium Bacillus subtilis (B. subtilis) have uncovered global changes in the transcriptome, altered mutagenic spectrum of DNA, and significant differences in cell susceptibility to fluoroquinolone antibiotics in spaceflight vs. ground controls. Fluoroquinolones target the enzymes DNA gyrase and topoisomerase IV, which are critical for maintaining DNA supercoiling as well as nucleoid compaction and conformation. Changes in DNA architecture affect many downstream genetic events such as: epigenetic DNA modifications (i.e., the methylome); the location of mutagenic hotspots in DNA; and global gene transcription patterns (i.e., the transcriptome). In turn, such alterations in global gene expression may lead to phenotypic changes observed in spaceflight. Analyses of the data from our prior missions has led us to formulate the following working hypothesis that can be tested experimentally using the BRIC-B. subtilis experimental system: Hypothesis: Exposure to spaceflight microgravity leads to changes in nucleoid architecture, DNA supercoiling, methylation, global transcription patterns, and resulting phenotypes. 2) Type of investigation: ISS with ground controls. 3) Specific aims: Using the B. subtilis model system, compare global: (i) 3-dimensional nucleoid architecture, (ii) DNA supercoiling, (iii) DNA methylation, and (iv) transcriptomes in spaceflight vs. matched ground controls. 4) Experimental plan: Experiments will utilize BRIC-Petri Dish Fixation Unit (PDFU) hardware modified with newly designed inserts mitigating the confounding effect of microgravity on liquid geometry. Four global parameters will be measured: (i) 3-dimensional nucleoid architecture using chromosome conformation capture (Hi-C); (ii) DNA supercoiling using 2-dimensional chloroquine-agarose gel electrophoresis; (iii) DNA methylation using Methyl-seq; and (iv) transcriptomes using RNA-seq. Results will probe at the molecular level the effect of spaceflight microgravity on four fundamental biological processes and their interactions. 5) NASA Relevance: This proposal directly addresses the Microbiology and the Cell and Molecular Biology Elements of the ROSBio-2018 Omnibus announcement and the NASA Strategic Plan Goals, especially Objective 1.2. The proposal directly addresses NASA's Space Biology Science Plan for 2016-2025, most particularly the overarching Microbiology Element Goal "to determine the effect of spaceflight on microbial life, processes, and community dynamics," specifically: 1) MB-1: What underlying genetic, molecular and biochemical processes are influenced by the space environment? (P2); 2) MB-2: What systems biology mechanisms and pathways for microbial life are responsive to spaceflight conditions? (P1, P2); 3) MB-4: What are the mechanisms that effect changes such as the altered virulence or altered drug resistance observed in some organisms during spaceflight? (P1, P2, P3); and 4) MB-5: How does the spaceflight environment influence microbial reproduction, growth, and physiology? (P1, P2).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of the project is to understand at a fundamental (molecular) level how bacteria sense and respond to the spaceflight environment. At present it is known that the phenotypic behavior of microbes is altered when confronted with spaceflight microgravity, but to date the underlying mechanism(s) involved have remained obscure. The project is to measure four fundamental molecular processes in cells of the model bacterium Bacillus subtilis: three-dimensional nucleoid architecture, DNA supercoiling, DNA methylation, and transcriptomes. These processes will be measured and compared in spaceflight (FL) vs. matched ground control (GC) samples. Deeper understanding of the molecular responses of bacteria to prolonged spaceflight has important implications for understanding disease processes and performance of microbes in bioreactors both in space and on Earth.
Task Progress:	November 2023 Update: Dr. Kelly Rice became Principal Investigator (PI) for this project August 21, 2023. For current reporting, please see Grant #80NSSC21K0319 ("Alteration of Bacillus subtilis DNA Architecture in Space: Global Effects on DNA Supercoiling, Methylation, and the Transcriptome") under PI Rice. (Ed., 11/20/23). ---------- NASA Grant 80NSSC21K0319, "Alterations in Bacillus subtilis DNA architecture in space: global effects on DNA supercoiling, methylation, and the transcriptome" Progress Report, FY23 (12/03/2021 to 12/02/2022) The Specific Aims of this project were to: Using the B. subtilis model system in spaceflight (FL) vs. matched ground controls (GC), compare global: A. 3-dimensional nucleoid architecture, B. DNA supercoiling, C. DNA methylation, and D. transcriptomes. Flight Hardware update. We designed, built, and tested new Petri dish inserts intended to keep liquid in the bottom of the dish regardless of gravity or its absence. The new dishes proved to function as designed in 2-second drop tower tests and were verified to be biocompatible with B. subtilis. In preliminary experiments, we have shown that under simulated "ISS" growth conditions, a culture grown in a single PDFU containing a modified Petri dish insert yields a final population of ~109 cells, sufficient for all downstream experiments. A manuscript describing the hardware and pre-flight experiments (Nicholson et al., Front. Space Technol. 2: 797518. 2021) was published in early FY22 (December 22, 2021). Specific Aim A was removed from the experiment. We originally proposed to measure the three-dimensional nucleoid architecture of B. subtilis cells grown either in FL or GC samples using a high-definition chromosome capture (Hi-C) technique in collaboration with the laboratory of Dr. Xindan Wang at Indiana University. The procedure required fixation of cells in 10% (v/v) formaldehyde, which triggered a safety review by NASA due to the possibility of release of formaldehyde fumes into the ISS interior in the unlikely event of a leak. Before the safety review was concluded, we discovered that cells were killed by formaldehyde fumes inside the PDFU, although no leakage was detected outside the hardware. Consultation with NASA concluded that this experiment would not fly as designed. Consequently, the remaining PDFUs were reassigned to increase the number of samples allocated to Specific Aims B, C, and D. Flight Update: Our spaceflight experiment to the ISS, designated BRIC-26, was tentatively scheduled for launch on SpaceX CRS-25 in late May 2022. However, due to several launch delays, it was finally launched on SpaceX CRS-26

	on November 26, 2022. The BRIC-26 payload was successfully actuated, incubated, and stored in the onboard -80°C MELFI freezer for return to Earth. As of the reporting date, the payload was still in the freezer on the ISS.
Bibliography Type:	Description: (Last Updated: 07/25/2025)